

HANGZHOU, China

WMO: 584570

Lat: 30.2259N

Lon: 120.1649E

Elev: 141

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.5	9.0	8.8	38.8	14.1	11.3	39.1	16.9	40.8	14.8	41.1	4.6	320	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	98.4	79.7	96.4	79.5	93.9	79.2	82.7	91.4	81.7	90.4	80.7	89.4	6.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	160.3	86.4	79.2	152.5	85.8	78.7	149.8	85.5	46.9	92.0	45.7	90.6	44.7	88.7	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.7	13.4	11.7	24.5	101.6	2.9	2.1	22.4	103.1	20.7	104.4	19.0	105.6	16.9	107.1		
			22.1	84.3	2.9	1.4	20.0	85.3	18.3	86.1	16.7	86.8	14.6	87.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	41.3	44.7	52.7	63.2	72.2	77.7	85.8	84.7	76.8	67.4	56.4	45.5	(6)
(7)		DBStd	16.26	5.98	7.04	7.44	6.79	5.15	5.12	5.01	4.59	5.45	5.19	6.66	6.10	(7)
(8)		HDD50	692	277	177	54	2	0	0	0	0	0	19	163		(8)
(9)		HDD65	2712	733	569	386	112	5	0	0	0	0	34	269	604	(9)
(10)		CDD50	5857	9	29	139	397	688	832	1109	1077	805	539	209	24	(10)
(11)		CDD65	2402	0	0	6	58	228	382	644	612	355	108	9	0	(11)
(12)		CDH74	24495	1	10	66	421	1560	3169	8437	7467	2874	446	44	0	(12)
(13)		CDH80	11056	1	1	13	115	502	1187	4555	3642	955	81	4	0	(13)
(14)	Wind	WSAvg	5.2	4.9	5.3	5.6	5.4	5.2	4.9	5.4	5.7	5.5	5.0	4.7	4.9	(14)
(15)	Precipitation	PrecAvg	54.3	2.4	3.3	4.7	5.1	6.3	7.2	5.0	5.8	6.7	3.4	2.4	1.9	(15)
(16)		PrecMax	81.2	5.6	10.6	12.8	8.7	13.6	17.4	11.7	12.9	16.1	11.6	9.7	5.5	(16)
(17)		PrecMin	37.6	0.0	0.5	1.3	1.7	3.3	1.9	1.1	0.5	1.5	0.0	0.1	0.0	(17)
(18)		PrecStd	9.2	1.5	2.1	2.6	1.8	2.3	3.5	2.8	3.1	3.8	2.7	2.0	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	73.8	81.0	88.2	93.1	96.8	101.8	100.7	95.4	86.3	78.9	66.0	(19)
(20)		0.4%	MCWB	54.5	58.9	62.5	67.9	72.4	79.4	79.3	78.8	79.6	72.1	64.8	55.7	(20)
(21)		2%	DB	58.6	65.3	74.3	83.3	89.3	93.2	98.9	97.9	91.6	82.2	73.6	61.1	(21)
(22)		2%	MCWB	49.4	55.1	59.6	66.3	69.5	78.4	79.7	79.6	78.0	69.1	62.0	53.0	(22)
(23)		5%	DB	53.6	59.5	69.6	78.7	85.7	89.7	97.0	95.3	88.1	78.7	69.7	57.6	(23)
(24)		5%	MCWB	47.2	51.0	57.3	64.6	69.5	75.9	79.5	79.7	76.8	67.8	61.4	50.8	(24)
(25)		10%	DB	50.0	55.1	64.6	75.0	82.2	86.4	95.1	93.2	85.6	75.6	66.3	55.1	(25)
(26)		10%	MCWB	45.2	48.4	55.6	62.5	68.2	75.2	79.4	79.6	75.3	65.6	59.8	49.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	61.2	66.4	73.0	77.8	81.9	84.1	84.0	82.8	76.4	69.5	59.5	(27)
(28)		0.4%	MCDWB	64.5	70.2	74.6	80.9	87.6	90.2	93.4	92.9	89.9	82.3	73.6	62.4	(28)
(29)		2%	WB	51.8	56.5	62.4	70.2	75.2	80.1	82.8	82.5	80.5	73.1	66.0	55.5	(29)
(30)		2%	MCDWB	55.8	63.6	69.7	76.9	82.4	88.4	92.0	91.0	87.5	77.8	70.2	58.3	(30)
(31)		5%	WB	48.4	52.5	59.4	67.8	73.4	78.8	81.8	81.5	78.8	70.7	63.5	52.8	(31)
(32)		5%	MCDWB	51.8	58.0	66.9	74.9	79.9	86.6	91.1	89.8	85.1	75.2	67.2	56.1	(32)
(33)		10%	WB	45.8	49.6	56.7	65.1	71.6	77.4	80.9	80.6	77.3	68.8	60.8	50.4	(33)
(34)		10%	MCDWB	48.8	53.9	63.3	72.4	77.5	84.1	90.2	88.8	83.4	73.4	65.0	53.6	(34)
(35)	Mean Daily Temperature Range		MDBR	12.0	13.1	15.0	16.3	15.7	13.1	14.8	13.9	13.1	14.2	14.3	13.5	(35)
(36)		5% DB	MCDBR	20.1	23.5	24.2	23.8	22.0	18.4	17.7	17.5	16.4	18.2	20.0	19.7	(36)
(37)		5% DB	MCWBR	12.2	13.2	12.1	10.6	9.1	7.3	6.1	6.3	6.6	7.7	10.2	11.6	(37)
(38)		5% WB	MCDBR	16.5	20.0	20.3	18.5	16.6	15.5	16.5	15.8	14.5	13.7	14.2	13.9	(38)
(39)		5% WB	MCWBR	11.3	12.5	11.9	10.5	8.9	7.2	6.8	6.4	6.6	7.6	8.8	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.608	0.649	0.687	0.692	0.665	0.691	0.642	0.667	0.657	0.632	0.611	0.583	(40)	
(41)		taud	1.647	1.586	1.550	1.591	1.684	1.671	1.799	1.724	1.714	1.694	1.689	1.698	(41)	
(42)		Ebn at Noon	188	196	202	209	216	210	220	212	207	198	186	186	(42)	
(43)		Edh at Noon	68	78	86	85	78	79	69	74	72	69	64	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	677	787	1092	1301	1407	1262	1665	1468	1170	1026	760	694	(44)	
(45)		RadStd	113	182	91	157	168	192	200	190	121	157	110	128	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.01	N/A	N/A	N/A	-1.08	-1.55	N/A	-149	+274	+153	(46)
(47)	Regional (14 neighbors)	+0.58	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+180	+117	(47)

Nomenclature: See separate page